

Feature Engineering, Model Optimization & Performance Comparison

Maincrafts Technology AIML Internship - Task 2

Introduction: This report summarizes an enhanced house price prediction workflow using the California Housing dataset. The project demonstrates feature scaling, multiple regression models, model evaluation, comparison, visualization, and final model persistence.

Objective: Compare Linear Regression, Ridge Regression, and Decision Tree Regressor models using RMSE and R2 Score, then select the best model using highest R2 Score and lowest RMSE as the tie-breaker.

Dataset: 20,640 records, 8 input features, target variable HousePrice. Features used: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude.

Feature Engineering: The data was inspected for missing values, separated into features and target, and scaled using StandardScaler before the 80/20 train-test split.

Models Evaluated:

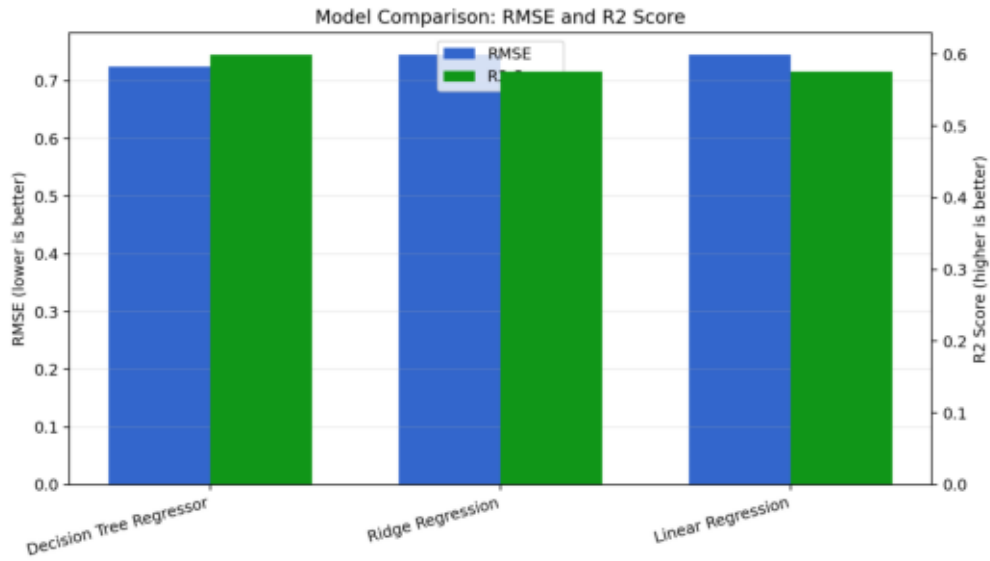
- Decision Tree Regressor: RMSE=0.7242, R2=0.5997
- Ridge Regression: RMSE=0.7456, R2=0.5758
- Linear Regression: RMSE=0.7456, R2=0.5758

Best Model: Decision Tree Regressor was selected with RMSE=0.7242 and R2=0.5997.

Conclusion: The selected model achieved the strongest test-set explanatory performance under the required selection rule. The workflow is reproducible and generates saved metrics, comparison tables, sample predictions, plots, and a reusable model bundle.

Task 2 Visualizations

Model Comparison



Actual vs Predicted



Feature Importance

